White Paper

ASD-STE100 Simplified Technical English and Artificial Intelligence (AI)

Exploring the impact of AI on ASD-STE100 content creation, translation, and technical documentation processes

Introduction

The purpose of this white paper is to clarify the position of the ASD Simplified Technical English Maintenance Group (STEMG) on the use of Artificial Intelligence (AI) in relation to ASD-STE100 Simplified Technical English (STE).

The rapid emergence of AI, particularly Large Language Models (LLMs), has significantly changed the creation, management, and translation of technical content. These technologies can process large volumes of data, support terminology consistency, and accelerate drafting and localization. Organizations now operate in a context where human expertise and AI capabilities must be integrated to maintain accuracy, standard compliance, and content quality.

For the STE community, this shift introduces both opportunities and risks, alongside the need for structured guidance. AI and LLMs can support faster content production, improve terminology control, and facilitate multilingual consistency. Conversely, they can compromise controlled natural language discipline, reduce traceability of content decisions, and introduce undetected inaccuracies.

The primary role of the STEMG is to develop and maintain the STE standard. It does not engage in producing AI software or tools and does not endorse any of them. However, ongoing monitoring is required to ensure that STE principles are always maintained and preserved. In all cases, the standard takes priority, and AI adoption must support, not override, STE requirements, preserving clarity, safety, and controlled natural language discipline.

The Role of AI in Technical Content Creation

AI and LLMs are already extensively used in drafting, translating, and managing technical content.

Potential benefits include:

- Increased efficiency and productivity
- Improved consistency across multilingual documentation
- Automated checks for STE compliance, but currently with varying levels of accuracy

However, challenges remain:

- Variable reliability and factual accuracy of AI-generated content
- Potential compromise of terminology control
- Limited traceability of content sources

For the STE community, AI tools can support STE-compliant writing and translation but cannot replace human oversight. Continuous intervention is necessary to ensure that STE principles remain the primary reference in all technical communication activities.

Key Ethical and Regulatory Framework

AI deployment in technical communication is subject to ethical and regulatory considerations:

- **EU AI Act:** Introduces a risk-based approach, classifying AI applications according to potential impact, especially for safety, reliability, or compliance.
- **UNESCO AI Ethics Principles:** Emphasize transparency, accountability, and the need for human oversight.
- **Other frameworks:** ISO standards on AI and OECD (Organisation for Economic Co-operation and Development) recommendations provide additional guidance on trustworthy AI deployment.

For STE, these frameworks underscore the importance of compliance and auditability in AI-assisted workflows, ensuring that AI use always aligns with the STE standard and remains accountable to human experts.

Risks and Safeguards

AI introduces specific risks in technical communication that require careful management:

- **Transparency and explainability:** Users must understand how AI outputs are generated.
- **Accountability:** Responsibility remains with the human author or organization.
- **Data confidentiality:** Protect proprietary and sensitive technical information.
- **Reliability and troubleshooting risks:** AI performance may be limited in safety-critical scenarios or with non-standard content.
- **Legal and reputational implications:** Disclaimers and clear boundaries of AI use are essential.

These safeguards help ensure that AI is applied responsibly without compromising STE principles.

STEMG Actions and Recommendations

The STEMG has already taken steps to protect the STE standard, including extending on behalf of ASD the non-endorsement policy to AI tools, consistent with the existing software and training practices. Recommended future actions include:

- **Maintain open dialogue** with AI developers and stakeholders to stay informed on technology trends.
- **Encourage proprietary or in-house AI tools** where confidentiality is critical.
- **Develop benchmarks or evaluation frameworks** to assess AI performance in STE-compliant writing and translation.
- **Monitor regulations, ethics, and best practices** to ensure STE principles are upheld.

Guidance for STE Users and Organizations

Responsible use of AI in STE environments requires clear guidance:

- **Supportive use only:** AI should assist, not replace, human authors.
- **Disclaimers:** Indicate when AI-assisted content is used, with scope and limitations.
- **Internal policies:** Define rules for data handling, model validation, and quality assurance.
- **Safe and unsafe practices:** Provide examples of compliant AI-assisted workflows and practices to avoid.

Obeying these recommendations allows organizations and users to leverage AI benefits while keeping STE principles as the primary reference.

Conclusions

AI will not replace human authors, but it is reshaping technical writing.

From drafting and terminology management to revision and quality assurance, AI is becoming an integral part of the technical writing process. The key challenge is no longer adoption, but responsible use: maintaining transparency, accountability, and strict adherence to STE principles and rules. By applying clear official policies, maintaining human oversight, and prioritizing the standard, organizations can benefit from AI's efficiency and consistency while safeguarding accuracy, safety, and professional integrity.

The STEMG and STEMG AITT (Artificial Intelligence Task Team),

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